



The WMO precipitation measurement intercomparisons

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ABSTRACT

This paper presents the results of international precipitation measurement intercomparisons organized by the World Meteorological Organization between 1955 and 2008. They have focussed on the impact of wind, evaporation and other systematic error sources on rain, snow and precipitation intensity measurement using standard gauges and different types of tipping-bucket and weighing gauges. The application of the results of WMO intercomparisons represents the state-of-the-art of precipitation measurements and the ultimate step to increase the reliability of point precipitation measurements and hydrological studies. It indicates that missing correction application makes any intercomparison measurements using different precipitation gauges 50 years behind the age and in any case, their results questionable.

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1. Introduction

Precipitation gauges of a different construction and elevation height installed at the same site near to each other frequently measure different precipitation amounts. The reason is the systematic error of precipitation measurement, consisting of losses due to evaporation and wetting and the most importantly wind-induced losses, which can vary from gauge to gauge. The latter is due to the elevated body of the gauge, which is an obstacle to the wind, as well as to its aerodynamical properties. In addition there are other error sources, particularly for precipitation recording gauges, which arise from the different mechanical and electronic equipment as well as the heating used by different types of gauges to record liquid and solid precipitation in small time intervals.

To eliminate these effects, precipitation gauge measurements have to be checked and corrected as recommended by the World

Meteorological Organization (WMO). For this aim the WMO established reference standards (Fig. 1) and organized to-date four international intercomparison studies of precipitation gauge measurements to derive correction techniques (Table 1). In these activities, the community of the best scientists in the world supported the WMO, particularly from the former Soviet Union, where the most extensive investigations have been carried out prior to the WMO intercomparisons. In the course of intercomparisons the fundamentals of the accuracy of precipitation measurements were defined and corrections of systematic measurement errors developed. This confirms the WMO leading role in the field. It considerably contributed to the solution of the basic problems of accuracy of precipitation measurement.

This paper provides an overview of the results of these international intercomparisons. The need for the calibration and the application of the correction of point precipitation measurements to increase their reliability is underlined. In the light of still published studies of suspicious quality on the topic, using uncorrected precipitation measurements and neglecting the state-of-the-art WMO intercomparison results, this statement is of a particular importance. In some studies the relevant literature is not mentioned or mentioned but not considered and the WMO recommendations were not followed. There is no reason to repeat the studies with no progress at all and with the same content again and again and present, occasionally, the wrong

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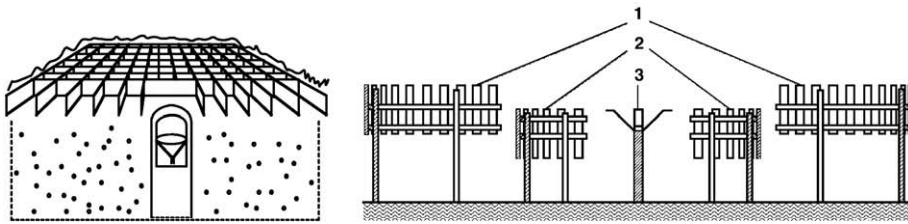


Fig. 1. Standard reference gauges as used during the second and third WMO international precipitation measurement intercomparisons: the pit gauge (left) and the Double-Fence International Reference DFIR (right). According to [Sevruk \(2004\)](#).

precipitation measurements and insufficient analysis as “challenges in obtaining reliable measurements of point rainfall” as still happens at present. Similarly, the manufacturers of precipitation gauges seem to have other problems than to apply intercomparison results to develop a proper gauge. The importance of objective laboratory and field intercomparison studies to evaluate the performance of precipitation gauges and to develop better gauges and correction techniques is stressed – they continue to be the main tool in precipitation measurement investigations.

2. WMO intercomparisons, 1960–2008

[Table 1](#) gives an outlook of the four WMO precipitation measurement intercomparisons including different gauge types as related to the legend, further the reference standard measurement used, the participating countries and the results obtained. [Fig. 1](#) shows the WMO reference standards

for liquid and solid precipitation measurements. [Sevruk \(2004\)](#) gives a complete history of the subject (in German). The state-of-the-art-report on corrections of wind-induced losses is described in [Sevruk \(2005\)](#). An excellent source of references concerning intercomparison measurements up to 1972 is the WMO Annotated Bibliography on Precipitation Measurement Instruments by [Rodda \(1973\)](#) and the review (in German) by [Sevruk \(1981\)](#).

2.1. Precipitation intercomparison, 1955–1975

The objective of the first intercomparison was to obtain reduction coefficients between the catches of various types of national gauges. The WMO and the International Association of Hydrological Sciences jointly organized it. The UK Snowdon gauge was chosen as the International Reference Precipitation Gauge (IRPG). It was elevated 1.0 m above the ground and equipped with the Alter wind shield. Such a gauge, however,

Table 1
WMO international precipitation measurement intercomparisons.

Comparison	I	II	III	IV
Subject	Precipitation	Rain	Snow	Rain intensity
Period	1955–1975	1972–1976	1986–1993	2004–2008
Purpose	Reduction coefficients between the catches of various types of national gauges	Rain catch differences between various types of national gauges and the pit gauge (Fig. 1). Correction procedures developed	Wind-induced error and standard correction procedures. (Wetting and evaporation losses considered)	Performance of different principles used to measure rainfall intensity (inherent mechanical and electronic errors)
Reference standard (Fig. 1)	Mk 2 gauge ^a elevated 1 m above the ground and equipped with the Alter wind shield	Pit gauge (Mk2) ^a installed in a pit, the orifice flush with ground and surrounded by anti-splash grid	Double-Fence Inter-national Reference, DFIR (Fig. 1) ^{bc}	Calibration in three independent laboratories in France, Italy and Netherlands for different rain intensities and field tests in Italy
Participants	Belgium, Czechoslovakia, Hungary, Israel, USA, Russia	Basic stations: 22 countries. Evaluation stations: Australia, Denmark, Finland, USA	Canada, China, Croatia, Denmark, Finland, Germany, Norway, Russia, Sweden, USA	12 tipping-bucket gauge models, 5 weighing gauges and 2 water level gauges, all from 15 countries ^d
Results	Non-conclusive	Wind-induced loss depends on wind speed, rain intensity and type of gauge. It amounts on average to 3% (up to 20%) and to 4–6% if wetting and evaporation losses are accounted for	Wind-induced loss depends on wind speed, temperature and type of gauge. Non-shielded gauges show greater losses as shielded ones (up to 80% vs. 40% for wind speed of 5 m/s and $t > -8$ °C)	Tipping-bucket gauges where no proper correction software was applied had larger errors than the weighing gauges. Problems of water storage in the funnel also occurred that could limit the range of measurements
Reference	Poncelet (1959) Struizer (1971)	Sevruk and Hamon (1984)	Goodison et al. (1998)	Lanza et al. (2005)

^a British Meteorological Office standard gauge of Snowdon type.

^b The Tretyakov gauge is the Russian standard gauge.

^c The diameter of inner fence is 4 m and of the outer fence is 12 m. The respective heights are 3 and 3.5 m above ground ([Fig. 1](#)). The Tretyakov gauge without fence is the secondary standard.

^d Australia, Austria, Canada, Czech Republic, Finland, France, Germany, India, Italy, Japan, Norway, Slovakia, Switzerland, UK, USA. The types of gauges are shown in [Sevruk and Klemm \(1989\)](#).

is still subject to a considerable extent to the wind field deformation and consequently, does not show the correct amount of precipitation. This could be why the first international intercomparison failed (Struzer, 1971) but its results (Poncelet, 1959) have been used to develop the first map of corrected global precipitation (UNESCO, 1978).

2.2. Rain intercomparison, 1972–1976

The objective of the second intercomparison was to evaluate wind correction factors for rainfall and to develop correction of systematic errors using the pit gauge surrounded by the anti-splash protection as the WMO standard reference (Fig. 1). Pit gauges are hardly affected by wind, and if corrected for wetting and evaporation losses they give reliable results.

The results showed that the point rainfall measurement is subject to the systematic wind-induced loss, which is on the order of 4–6% depending on the gauge type and the latitude and altitude of the gauge site. This error can be corrected using an empirical model based on meteorological variables such as wind speed and the intensity of precipitation (Sevruck and Hamon, 1984). More sophisticated correction method is based on the numerical simulation (Nespor and Sevruck, 1999).

2.3. Snow intercomparison, 1986–1993

The aim was to determine the wind-induced error of different shielded and unshielded national standard gauges and to derive correction procedures for solid and mixed precipitation measurements considering wetting and evaporation losses. From the numerous snowfall measurement techniques the precipitation gauges shielded by fences appeared to be the most promising. The Russian double fence was finally selected as the WMO reference standard (DFIR). It consists of the shielded Tretyakov gauge encircled by two octagonal lath-fences with a diameter of 4 and 12 m and respective heights of 3 and 3.5 m as shown in Fig. 1. The results for snow and mixed precipitation are given in Goodison et al. (1998). They are reproduced in Fig. 2 for five types of gauges, three out of them with wind shield and two without it. It is shown that due to the wind-induced losses

during snowfall unshielded gauges catch considerably less precipitation (P_m) than the shielded ones and the reference gauge DFIR (P_r). The ratio $P_m P_r^{-1}$ indicates that in the case of a wind speed of 6 m s^{-1} the unshielded Hellmann and USWB (US Weather Bureau) gauges catch only 20% of the “true” amount of snow and 50% of the mixed precipitation. The respective figures for the shielded gauges vary between 25% and 55% and 20% and 30% for the mixed precipitation.

2.4. Rain intensity intercomparison, 2004–2008

The main objectives of the WMO Laboratory Intercomparison of Rainfall Intensity Gauges were:

- to test the performance of different principles used to measure rainfall intensity under documented laboratory conditions,
- to define a standardized calibration procedure and
- to provide information on improving the homogeneity of rainfall time series considering high rainfall intensities.

The results were published by Lanza et al. (2005) and are shown in Fig. 3.

The methodology was based on the generation of a constant water flow within the range of operational use. The water was conveyed to the funnel of the gauge in order to simulate constant rainwater intensity. This value was considered as the reference and is indicated by the symbol “I reference” in Fig. 3. The flow was measured by weighing the water over a given period of time. The intensity measured by the gauge was indicated by the symbol “I measured”. The relative differences between the simulated and generated rainfall intensities were assumed to be the relative error, and they are shown in Fig. 3 for different types of gauges.

The results showed that the tipping-bucket rain gauges that were equipped with proper correction software provided a good quality of rainfall intensity measurements (Fig. 3). Three models (CAE, ETG and Yokogawa) showed very small errors while two models (Waterlog and Meteoservis MR3H) approached the 5% error limit. The gauges where no correction was applied had larger errors (Fig. 3). Particularly, the five models (Lamprecht, Casella, Paar, SIAP and MS) showed very poor performance with errors larger than the 5%

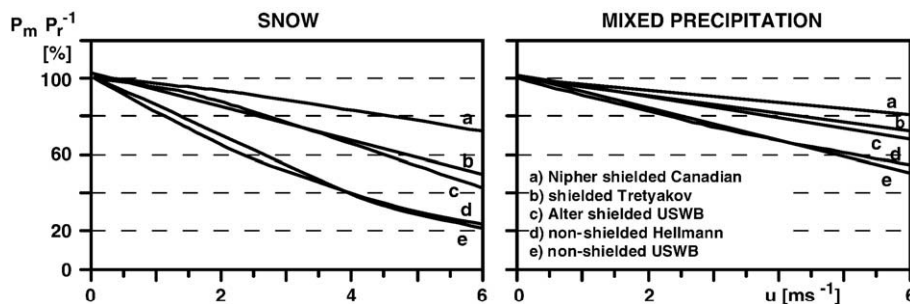


Fig. 2. The dependence of the ratio $P_m P_r^{-1}$ of measured precipitation P_m and the reference precipitation P_r , as measured using DFIR (Fig. 1), on the wind speed u and the use of a wind shield (gauges a; b; c) for solid (left) and mixed (right) precipitation, according to Goodison et al. (1998). The gauges d and e are non-shielded Hellmann and USWB gauges. Note that these unshielded gauges catch only 20% of the snow as measured by DFIR during a wind speed of 6 m s^{-1} . For the shielded gauges b) the Russian Tretyakov and c) the USWB gauge the catch amounts to 50% relative to the DFIR catch. The best performance is shown by the Canadian Nipher wind shield a).

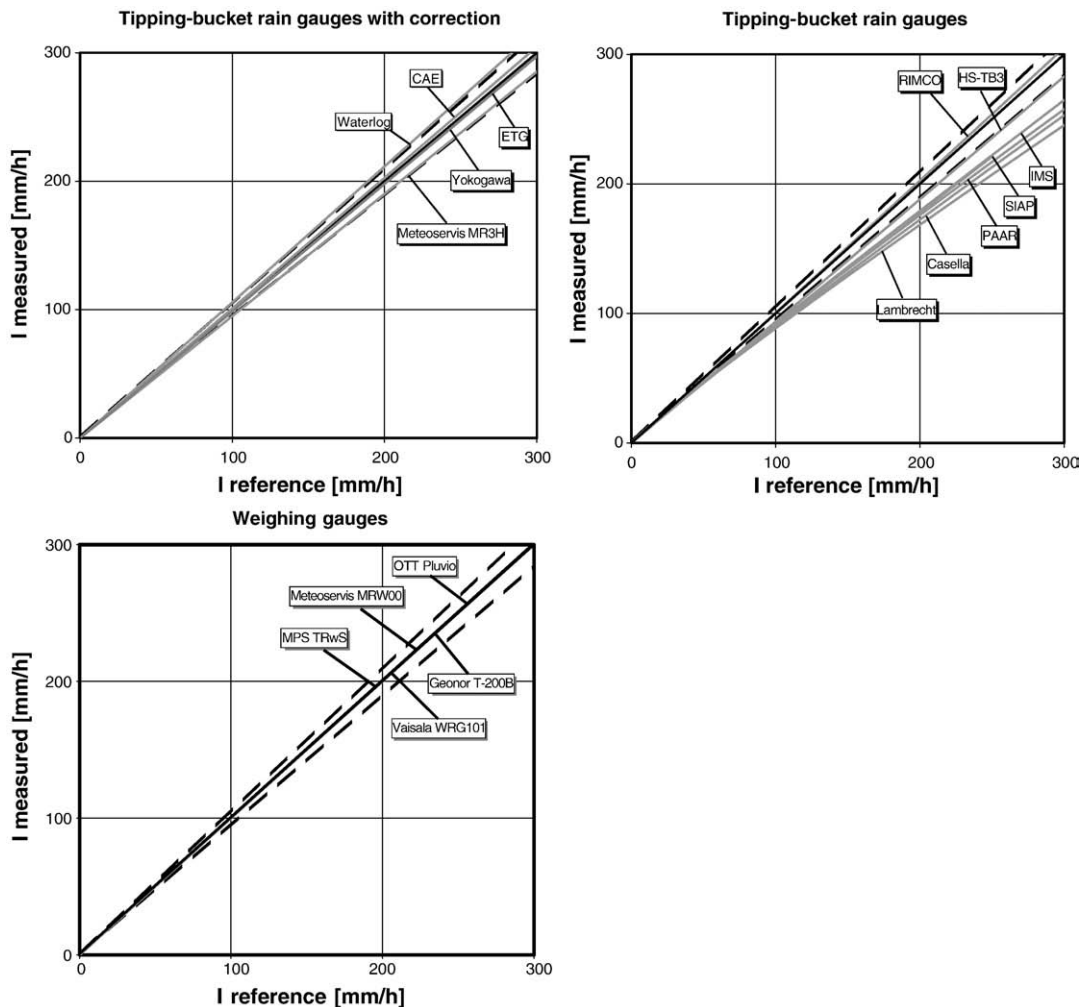


Fig. 3. Calibration curves for the five different tipping-bucket rain recording gauges with correction (top left), seven different tipping-bucket rain recording gauges without correction (top right), and five different weighing rain recording gauges (bottom left), according to Lanza et al. (2005). The symbol “I reference” indicates the simulated intensity and “I measured” the measured intensity by a particular gauge. Dashed lines show error bars of $\pm 5\%$ and insets the respective type of gauge.

limit. In some cases, problems of water storage in the funnel occurred that could limit the usable range for rain intensity measurement. The error is generally smaller for the electronic weighing gauges than for the tipping-bucket rain gauges (Fig. 3), provided the instrument is properly stabilized.

The measurement of rainfall intensity is affected by the response time of the acquisition system. Significant delays were observed in “sensing” time variation of the rainfall intensity. The delay is the result of the internal software, which is intended to filter the noise. Only one model of weighing gauges (Meteoservis) had a delay that met the WMO 1-minute rainfall intensity requirement.

The two other gauges (Alluvion, Canada, and Serosi, France) which used a conductivity measurement method for determining water level showed good performances. Siphoning problems with one of these gauges (Alluvion) limits its ability to measure a wide range of rainfall intensity. For the other one, a limitation was found related to the emptying mechanism where a 2 min delay was observed. These two gauges, potentially sensitive to water conductivity had no demonstrated problems.

2.5. Field intercomparison, 2007–2008

The results obtained during the IV. intercomparison can provide preliminary information to manufacturers and a first step selection criterion for the user. Therefore, it was necessary to proceed with the quality assessment procedure initiated in the laboratory by organizing a follow-up field intercomparison using pit gauges as the reference, as is carried-out at present in Vigna di Valle, Italy. (See: <http://www.wmo.int/pages/prog/www/IMOP/intercomparisons.html>). This allows continuity in the performance assessment procedure and result in the estimation of the overall operational error to be expected in the measurement of rain intensity in the field. The preliminary results are to be expected at the end of this year.

3. Summary

The WMO intercomparison measurements of point precipitation considerably contributed to the solution of crucial problems of precipitation measurement accuracy using

precipitation gauges. The reference gauges were developed and correction procedures derived. They show the need of application of corrections of systematic measurement errors to acquire more reliable precipitation data. The importance of intercomparison measurements is stressed by the fact that they are used to check the performance of precipitation gauges and to develop correction procedures for systematic error of precipitation measurement. Intercomparisons of different gauge types and measurement techniques continue to be the main tool in precipitation measurement investigations and development of better gauges.

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